

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

593000606 - Scientific Research Methodology

DEGREE PROGRAMME

59AJ - Master Universitario En Comunicaciones Inalámbricas

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

Index

Learning guide

1. Description.....	1
2. Faculty.....	1
3. Skills and learning outcomes	2
4. Brief description of the subject and syllabus.....	3
5. Schedule.....	5
6. Activities and assessment criteria.....	7
7. Teaching resources.....	10

1. Description

1.1. Subject details

Name of the subject	593000606 - Scientific Research Methodology
No of credits	3 ECTS
Type	Compulsory
Academic year of the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	59AJ - Master Universitario en Comunicaciones Inalámbricas
Centre	59 - E.T.S. De Ingeniería Y Sist. De Telecom.
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Marta Gil Barba	D8415	marta.gil.barba@upm.es	Sin horario. Meetings with prior appointment.
David Luengo Garcia (Subject coordinator)	A7011, D8201A	david.luengo@upm.es	Sin horario. Meetings with prior appointment.

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Skills and learning outcomes *

3.1. Skills to be learned

CEM13 - Adquirir un conocimiento instrumental mínimo que permita plantear formalmente un proyecto de investigación. / Acquire a minimum instrumental knowledge that allows to formally propose a research project.

CGI02 - Comprender el procedimiento, valor y límites del método científico, siendo capaz de identificar, localizar y obtener datos requeridos en un trabajo de investigación, de diseñar y guiar investigaciones analíticas, de modelado y experimentales, así como de evaluar datos de una manera crítica y extraer conclusiones. / Understand the procedure, value, and limits of the scientific method, being able to identify, locate and obtain data required in a research work, to design and guide analytical, modeling, and experimental investigations, as well as to critically evaluate data and extract conclusions.

CGI03 - Valorar la importancia de las fuentes documentales, manejarlas y buscar la información para el desarrollo de cualquier trabajo de investigación. / Assess the importance of documentary sources, manage them and search for information for the development of any research work.

CGI05 - Adquirir el conocimiento necesario sobre los mecanismos de financiación de la investigación y transferencia de la tecnología, y sobre la legislación vigente sobre protección de resultados. / Acquire the necessary knowledge about the mechanisms for financing research and technology transfer, and about current legislation on the protection of results.

UPM1 - Uso de la lengua inglesa / Use of the English language

UPM2 - Liderazgo de equipos / Team leadership

3.2. Learning outcomes

RA20 - Utilizar adecuadamente los recursos bibliográficos y bibliométricos disponibles/ Properly use of the available bibliographic and bibliometric resources

RA23 - Localizar convocatorias de ayudas para investigación y redactar adecuadamente las propuestas/ Find out calls for research grants and properly draft proposals

RA19 - Entender el proceso y las características de la actividad investigadora/ Understand the process and characteristics of the research activity

RA21 - Adquirir una actitud crítica y pragmática de las teorías sobre el conocimiento científico/ Acquire a critical and pragmatic attitude in relation to the theories about scientific knowledge

RA22 - Presentar y defender con rigor un trabajo de forma escrita y oral en inglés/ Present and rigorously defend a work in written and oral form in English

RA8 - Identificar el impacto social y económico de las comunicaciones móviles en un contexto global/ Identify the social and economic impact of mobile communications in a global context.

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

4. Brief description of the subject and syllabus

4.1. Brief description of the subject

This course is aimed at introducing students to concepts and practices of the scientific research methodology. During the course, students will demonstrate their understanding and competence through the development of the study design for their own research project.

4.2. Syllabus

1. Scientific research processes
 - 1.1. Philosophy of science
 - 1.2. Scientific research method
 - 1.3. The research process - stages of research
2. Bibliographic resources and bibliometrics
 - 2.1. Research digital ID (Orcid, Scopus, WoS, etc.)
 - 2.2. UPM's Scientific Portal
 - 2.3. Ingenio and other information sources
 - 2.4. References and bibliographic managers
3. Communication techniques
 - 3.1. Scientific and technical language
 - 3.2. Elaboration of scientific documents
 - 3.3. Oratory and communication skills
4. Scientific policy
 - 4.1. Introduction: ways of funding your research projects
 - 4.2. Calls for Spanish Government funding projects
 - 4.3. Calls for European Union funding projects
 - 4.4. Private research collaboration projects
5. Ethical aspects of scientific work
 - 5.1. The concept of ethics in research: ethical committees
 - 5.2. Invention, forgery and plagiarism
 - 5.3. Confidentiality, copyright and conflicts of interest

5. Schedule

5.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Course presentation & Scientific research processes Duration: 02:00 Scientific research processes Duration: 02:00			Classroom Assessment - Unit 1 Progressive assessment and Global Examination Presential Duration: 00:00
2	Bibliographic resources and bibliometrics Duration: 02:00 Bibliographic resources and bibliometrics Duration: 02:00			Classroom Assessment - Unit 2 Progressive assessment and Global Examination Presential Duration: 00:00 Final Work - Milestone 1 (Topic Selection: Title, Abstract & Keywords) Progressive assessment and Global Examination Not Presential Duration: 00:00
3	Bibliographic resources and bibliometrics Duration: 02:00 Communication Techniques Duration: 02:00			Classroom Assessment - Units 2 & 3 Progressive assessment and Global Examination Presential Duration: 00:00
4	Communication Techniques Duration: 02:00 Communication Techniques Duration: 02:00			Final Work - Milestone 2 (Problem Description + State of the Art) Progressive assessment and Global Examination Not Presential Duration: 00:00 Classroom Assessment - Unit 3 Progressive assessment and Global Examination Presential Duration: 00:00

5	Scientific Policy Duration: 02:00 Scientific Policy Duration: 02:00			Classroom Assessment - Unit 4 Progressive assessment and Global Examination Presential Duration: 00:00
6	Ethical Aspects of Scientific Work Duration: 02:00 Ethical Aspects of Scientific Work Duration: 02:00			Final Work - Milestone 3 (Budget, Workplan, Exploitation & Dissemination) Progressive assessment and Global Examination Not Presential Duration: 00:00 Classroom Assessment - Unit 5 Progressive assessment and Global Examination Presential Duration: 00:00
7	Final works presentation Duration: 02:00 Final works presentation Duration: 02:00			Submission of Final Work's Memory Progressive assessment and Global Examination Not Presential Duration: 00:00 Presentation of Final Works Group presentation in the classroom Progressive assessment and Global Examination Presential Duration: 02:00
8				
9				
10				
11				
12				
13				
14				
15				
16				
17				Global Written Exam (only for students failing the progressive evaluation) Global examination Presential Duration: 02:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

6. Activities and assessment criteria

6.1. Assessment activities

6.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
1	Classroom Assessment - Unit 1		Face-to-face	00:00	5%	4 / 10	CGI02 CGI03 UPM1
2	Classroom Assessment - Unit 2		Face-to-face	00:00	5%	4 / 10	
2	Final Work - Milestone 1 (Topic Selection: Title, Abstract & Keywords)		No Presential	00:00	5%	/ 10	UPM1 UPM2 CEM13
3	Classroom Assessment - Units 2 & 3		Face-to-face	00:00	5%	4 / 10	
4	Final Work - Milestone 2 (Problem Description + State of the Art)		No Presential	00:00	10%	/ 10	CGI02 CGI03 UPM1 UPM2 CEM13
4	Classroom Assessment - Unit 3		Face-to-face	00:00	5%	4 / 10	
5	Classroom Assessment - Unit 4		Face-to-face	00:00	5%	4 / 10	
6	Final Work - Milestone 3 (Budget, Workplan, Exploitation & Dissemination)		No Presential	00:00	15%	/ 10	CGI02 CGI03 UPM1 UPM2 CEM13
6	Classroom Assessment - Unit 5		Face-to-face	00:00	5%	4 / 10	
7	Submission of Final Work's Memory		No Presential	00:00	10%	4 / 10	
7	Presentation of Final Works	Group presentation in the classroom	Face-to-face	02:00	30%	4 / 10	CGI02 CGI03 CGI05 UPM1 UPM2 CEM13

6.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
1	Classroom Assessment - Unit 1		Face-to-face	00:00	5%	4 / 10	CGI02 CGI03 UPM1
2	Classroom Assessment - Unit 2		Face-to-face	00:00	5%	4 / 10	
2	Final Work - Milestone 1 (Topic Selection: Title, Abstract & Keywords)		No Presential	00:00	5%	/ 10	UPM1 UPM2 CEM13
3	Classroom Assessment - Units 2 & 3		Face-to-face	00:00	5%	4 / 10	
4	Final Work - Milestone 2 (Problem Description + State of the Art)		No Presential	00:00	10%	/ 10	CGI02 CGI03 UPM1 UPM2 CEM13
4	Classroom Assessment - Unit 3		Face-to-face	00:00	5%	4 / 10	
5	Classroom Assessment - Unit 4		Face-to-face	00:00	5%	4 / 10	
6	Final Work - Milestone 3 (Budget, Workplan, Exploitation & Dissemination)		No Presential	00:00	15%	/ 10	CGI02 CGI03 UPM1 UPM2 CEM13
6	Classroom Assessment - Unit 5		Face-to-face	00:00	5%	4 / 10	
7	Submission of Final Work's Memory		No Presential	00:00	10%	4 / 10	
7	Presentation of Final Works	Group presentation in the classroom	Face-to-face	02:00	30%	4 / 10	CGI02 CGI03 CGI05 UPM1 UPM2 CEM13
17	Global Written Exam (only for students failing the progressive evaluation)		Face-to-face	02:00	30%	/ 10	CGI02 CGI03 CGI05 UPM1 CEM13

6.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Written Evaluacion (Units 1-5)		Face-to-face	01:00	30%	4 / 10	CGI02 CGI03 CGI05 UPM1 CEM13
Final Work's Memory		Face-to-face	00:00	40%	4 / 10	CGI02 CGI03 CGI05 UPM1 UPM2 CEM13
Final Work's Presentation		Face-to-face	00:30	30%	4 / 10	CGI02 CGI03 CGI05 UPM1 UPM2 CEM13

6.2. Assessment criteria

The progressive evaluation will consist of two parts:

- 1) Classroom assessment of each of the 5 units of the course (6 weeks, 5% per week).
- 2) A work to be delivered and presented at the end of the course (70%). Depending on the number of students enrolled in the course this work will be either individual or in groups of 2-3 students.

The first part can be recovered in the final and/or extraordinary exam by students failing to pass the course through a global written exam (30%) that will include questions from the 5 units of the course. The second part is not recoverable and **MUST** be done during the course.

7. Teaching resources

7.1. Teaching resources for the subject

Name	Type	Notes
Carl J. Sindermann, "Winning the games scientists play: strategies for enhancing your career in science", Perseus Publishing, 2001	Bibliography	
Leslie A. Olsen and Thomas N. Hucklin, "Technical writing and professional communication", McGraw-Hill, 1991 (2nd Ed).	Bibliography	
C. George Thomas, "Research Methodology and Scientific Writing", Springer, 2021 (2nd Ed).	Bibliography	
E. B. Wilson, "An Introduction to Scientific Research", Dover Publications, 2012.	Bibliography	
Uwem Essia, "Lecture Notes on Research Methodology (Books 1-5)", 2022	Bibliography	
On-Line Resources	Web resource	Journal and conference papers, scientific reports, on-line bibliographic resources, websites of different calls for projects, etc.